CSE400 – Fundamentals of Probability in Computing

Lecture 7: Expectation, CDFs, PDFs and Problem Solving

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1. Random Variables

A **random variable (RV)** is a function that maps outcomes of a random experiment to real numbers.

Random variables may be:

- **Discrete**
- **Continuous**

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2. Cumulative Distribution Function (CDF)

2.1 Definition

The **Cumulative Distribution Function (CDF)** of a random variable $X$is defined as:

$$F_X(x) = \Pr(X \leq x)$$

This definition applies to **both discrete and continuous random variables**.

2.2 Properties of the CDF

For any random variable X, the CDF satisfies:

1. **Non-decreasing property**

$$F_X(x_1) \leq F_X(x_2), \text{for } x_1 < x_2$$

2. **Right-continuity**

$$\lim_{x \to a^+} F_X(x) = F_X(a)$$

3. **Boundary conditions**

$$\lim_{x \to -\infty} F_X(x) = 0$$
$$\lim_{x \to \infty} F_X(x) = 1$$

4. **Probability of an interval**

$$\Pr(a < X \leq b) = F_X(b) - F_X(a)$$

2.3 Example of a CDF

An example CDF is typically illustrated using a **step function** for a discrete RV or a **smooth curve** for a continuous RV. (The formula is stated without derivation.)

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3. Probability Density Function (PDF)

3.1 Definition

For a **continuous random variable**, the **Probability Density Function (PDF)** $f_X(x)$is defined as:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Conversely,

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$$

3.2 Properties of the PDF

1. **Non-negativity**

$$f_X(x) \geq 0 \forall x$$

2. **Total probability**

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1$$

3. **Probability over an interval**

$$\Pr(a \leq X \leq b) = \int_{a}^{b} f_X(x)\, dx$$

3.3 Example of a PDF

An example PDF is illustrated graphically.
(The functional form is stated directly, without derivation.)

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4. Expectation of Random Variables

4.1 Definition of Expectation

The **expectation (mean)** of a random variable $X$is defined as:

- **Discrete RV**

$$\mathbb{E}[X] = \sum_{x} x \Pr(X = x)$$

- **Continuous RV**

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \; f_X(x) \, dx$$

4.2 Example of Expectation

Expectation is computed by directly applying the above formulas to the given PMF or PDF.
(No derivation is provided.)

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5. Expectation of a Function of a Random Variable

Let $Y = g(X)$.

- **Discrete case**

$$\mathbb{E}[g(X)] = \sum_{x} g(x) \Pr(X = x)$$

- **Continuous case**

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) \; f_X(x) \, dx$$

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6. Linear Operations with Expectation

Expectation satisfies **linearity**:

$$\mathbb{E}[aX + b] = a \, \mathbb{E}[X] + b$$

For two random variables $X$and Y:

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

This property holds **regardless of dependence**.

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7. Moments of Random Variables

7.1 nth Moment

The n**th moment** of Xabout the origin is defined as:

$$\mathbb{E}[X^n]$$

7.2 Central Moments

The n**th central moment** is defined as:

$$\mathbb{E}[(X-\mu)^n]$$

where $\mu = \mathbb{E}[X]$.

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8. Variance

The **variance** of a random variable $X$is:

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$$

Equivalent form:

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

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9. Skewness

Skewness is the **third central moment**, measuring asymmetry:

$$\mathbb{E}[(X-\mu)^3]$$

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10. Kurtosis

Kurtosis is the **fourth central moment**, measuring peakedness:

$$\mathbb{E}[(X-\mu)^4]$$